



NEXUS ENGINE

Deep Packet Inspection — Explained Simply

“See Every Packet. Trust Every Connection.”

Project Documentation | Real-Time Network Visibility & Threat Blocking Engine

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● IN ONE LINE

Nexus Engine watches raw network traffic in real time, figures out which website or app every connection belongs to — even when it is encrypted — and instantly allows or blocks it based on rules a network admin sets.

1. Who Is Nexus Engine For?

Nexus Engine is not an app a regular person installs on their phone. It is a control-room tool built for the people who are responsible for an entire network — deciding what traffic is allowed to pass through a company, a school, or an internet service.

Who	Why They'd Use Nexus Engine
Internet Providers (ISPs)	Manage which apps get priority, or block certain traffic types
Company IT / Security Teams	Stop employees from using unsafe or distracting apps at work
School Network Admins	Block social media or streaming sites on the school Wi-Fi
Security Researchers	Study how traffic behaves and spot suspicious activity in real time

● THINK OF IT LIKE THIS

Nexus Engine is the security guard's scanner at a building entrance — not something the visitor operates, but something the building's security team uses to decide who and what gets let in. The end user is a network administrator or security analyst, not an everyday web browser.

2. What Is Nexus Engine? (In Simple Words)

Every time a device opens a website or an app, it sends and receives tiny pieces of data called packets — little envelopes constantly flying back and forth between the device and the internet. A normal firewall only reads the outside of the envelope, the address. Nexus Engine looks a little bit inside the envelope too — enough to work out which website or app it belongs to, even when the contents are locked (encrypted).

Nexus Engine does three things with every flow it sees:

- **Identifies** which website/app each packet belongs to (YouTube, Facebook, QUIC, and more)
- **Decides** whether to allow or block it, based on rules the admin has set up
- **Reports** what it saw — live, on a dashboard — how much traffic, from which apps, how much got blocked

● THINK OF IT LIKE THIS

Imagine a mail-sorting centre. A basic sorter only reads the street address on an envelope. Nexus Engine is a smarter sorter — it peeks at the very first line of the letter, before it gets sealed, to figure out who it's really for, even if the rest of the letter is locked in a safe.

3. Why Does This Even Matter?

Here is the surprising part: even when a website uses HTTPS (the padlock icon in the browser), the very first message a browser sends — before anything gets encrypted — still contains the website's name in plain, readable text. This message is called the **TLS Client Hello**, and the readable part inside it is called **SNI (Server Name Indication)**.

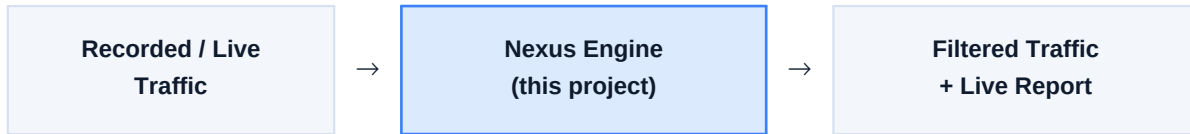
● **GOOD TO KNOW**

So even 'secure' HTTPS traffic leaks the domain name for a brief moment. Nexus Engine's biggest trick is catching that one moment, before the rest of the conversation locks up.

This is exactly why the technology matters in the real world: it lets a network operator know what is being accessed, without needing to break the encryption or read anyone's actual messages, photos, or passwords. Visibility without spying — that is the whole point.

4. How Does It Work? — Step by Step

At the highest level, here is the entire journey in one line:



Traffic arrives as a stream of packets, timestamped one after another. Every single packet is like a set of nested envelopes, one inside another:

- Ethernet Header — like the delivery truck's license plate
- IP Header — like the street address
- TCP/UDP Header — like the apartment / room number
- Payload / Actual Data — like the letter inside the envelope

Nexus Engine peels these layers open, one at a time, to understand: who sent it, who it is going to, and — most importantly — what website or app it belongs to.

5. The Journey of One Packet

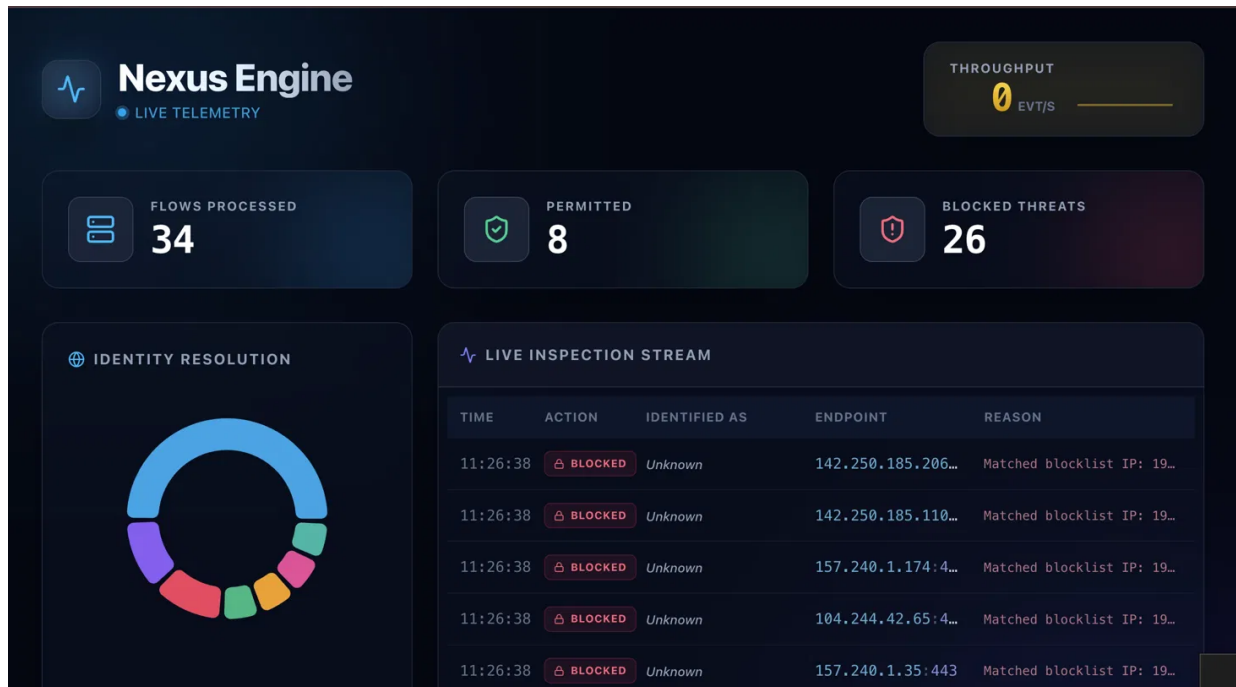
Step	What Happens
Step 1 — Read the Stream	The engine reads packets one by one, in the order they arrive or were captured.
Step 2 — Unwrap the Headers	It reads the Ethernet header, then the IP header, then the TCP/UDP header — like peeling back layers of an envelope.
Step 3 — Identify the Conversation	It groups the packet with others sharing the same 'Five-Tuple' — source, destination, ports and protocol — recognising which ongoing 'call' this packet belongs to.
Step 4 — Peek at the First Handshake	If it's the first message of an HTTPS connection, the engine looks for the SNI (website name) hidden in plain text.
Step 5 — Match Against the Rules	It checks: is this IP blocked? Is this app blocked? Does this domain match a blocked keyword?
Step 6 — Allow or Block	If nothing matches a rule, the flow is permitted. If something matches, it is blocked — and every future packet in that same conversation is blocked too.
Step 7 — Update the Live Report	The dashboard updates instantly: flows processed, permitted count, blocked-threat count, and identity breakdown.

● THINK OF IT LIKE THIS

It's like a bouncer at a club checking IDs. Once the bouncer realises 'this person is on the banned list,' they don't just turn them away once — they remember that face and turn them away every single time they try to come back in that same night.

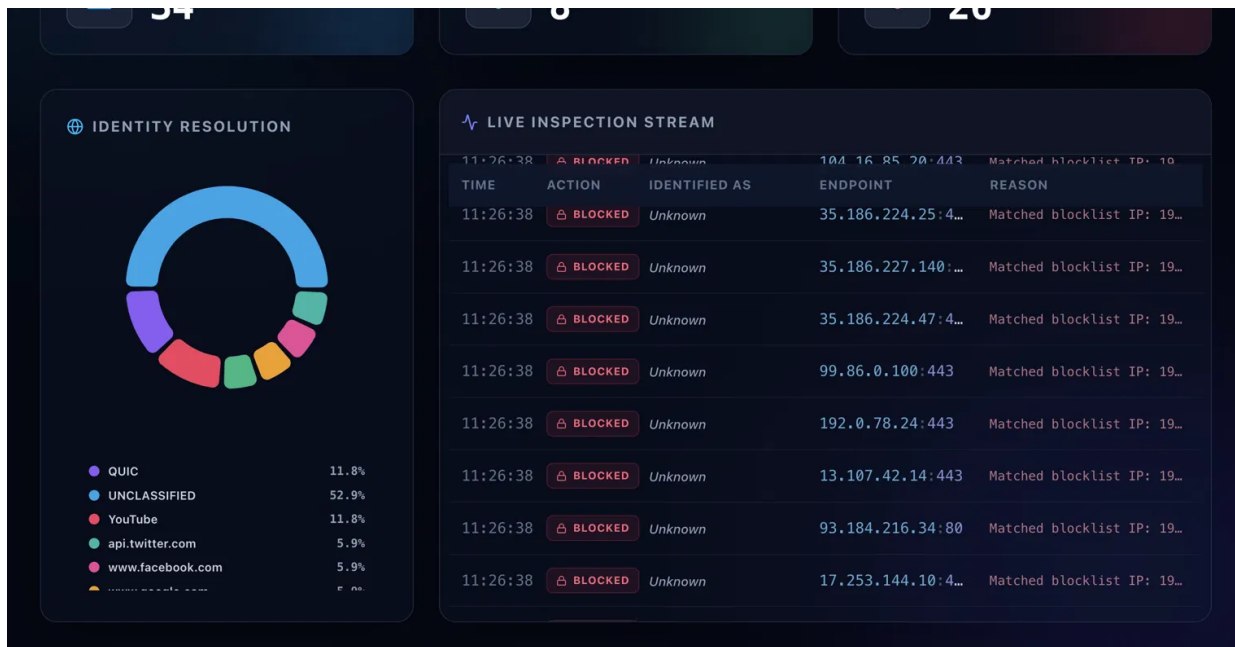
6. Inside the Live Dashboard

Everything Nexus Engine sees is surfaced live on its dashboard, so an administrator never has to read raw logs. Here is what a live session looks like while the engine is inspecting traffic:



Live telemetry: flows processed, permitted vs. blocked counts, identity resolution, and the live inspection stream showing each blocked endpoint and the matching reason.

At the top, the **Flows Processed**, **Permitted**, and **Blocked Threats** counters update in real time as new packets arrive. The **Identity Resolution** donut chart shows what share of traffic belongs to each recognised app or protocol (QUIC, YouTube, Facebook, and so on) versus traffic that remains unclassified. The **Live Inspection Stream** lists every action the engine takes — the timestamp, whether it was blocked, the endpoint IP:port involved, and the exact reason, such as a matched blocklist IP.



A closer look at Identity Resolution — the breakdown of QUIC, unclassified, YouTube, Facebook, and other traffic — alongside the continuously scrolling inspection stream.

This is the same idea as the DPI Engine's summary report, but turned into a living dashboard instead of a static end-of-run printout — an admin can watch decisions happen packet by packet, second by second.

7. How Blocking Decisions Are Made

A network admin sets up simple rules ahead of time. Nexus Engine checks every packet's conversation against these rules, in order:

Check	Example Rule	What Happens If It Matches
Is the source IP blocked?	Block 192.168.1.50	Drop everything from that device
Is the app blocked?	Block YouTube	Drop all YouTube traffic
Does the domain match?	Block anything with 'tiktok'	Drop that connection

Because decisions apply to the whole **conversation** (the five-tuple), not just a single packet, once a flow is flagged as 'Unknown / Matched blocklist IP', every remaining packet in that same conversation is blocked automatically — exactly what is visible scrolling through the Live Inspection Stream on the dashboard.

8. Under the Hood — Tools & Techniques

Technique	What It's Used For
Raw byte parsing & bit manipulation	Reading Ethernet / IP / TCP headers directly from memory
Network byte order conversion	Making sure numbers are read correctly across different systems
Hash maps	Quickly looking up which conversation a packet belongs to
Multi-threading	Running multiple worker threads at once for real-time speed
Mutexes & condition variables	Letting threads safely share data without corrupting it
Producer-consumer queues	Passing packets safely between Reader, Workers, and Writer
Consistent hashing	Making sure one conversation always goes to the same worker thread

● SPEED AT SCALE

Real networks produce huge amounts of traffic every second, so Nexus Engine splits work across a Reader, Load Balancer, Fast-Path Workers, and a Writer running in parallel — while making sure packets from the same conversation always land on the same worker, so no decision is ever made with only half the picture.

9. What You Get When You Run It

Output	What It Contains
Live Dashboard	Real-time flows processed, permitted vs. blocked counters, and identity breakdown
A Live Inspection Stream	A running, timestamped log of every allow/block decision and its reason
A Summary Report	How many packets total, how many blocked, and a breakdown by app
A List of Detected Identities	Every website / app / protocol the engine managed to recognise

● GOOD TO KNOW

Nothing about a person's actual messages, passwords, or private content is ever read or stored — only the destination name (like 'youtube.com') is identified, the same way a mail sorter reads an address without opening the envelope.

10. Key Takeaways

- Nexus Engine is a tool for network administrators, not everyday app users.
- It reads network traffic and figures out which website or app each connection belongs to — even when encrypted — by catching one small unencrypted moment at the very start of the connection.
- It groups packets into conversations so that a block/allow decision applies consistently to the whole flow.
- It surfaces everything live — counters, identity breakdown, and a scrolling inspection stream — instead of only a static end-of-run report.
- It is built for speed at real-world scale, using multi-threading and consistent hashing under the hood.

● THE BIG PICTURE

In one sentence: Nexus Engine is a smart, fast, live-watching sorting robot for internet traffic — it reads just enough of each envelope to know which app it belongs to, and hands network administrators the power to decide, instantly, what gets through.

“See Every Packet. Trust Every Connection.”